Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Information Technology

(Department of Electronics & Telecommunication)

Group No.: - 09

Project Report on

"Sign Language to speech conversion"

Domain: Deep Learning

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B. Tech Electronics & Telecommunication

Of

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Under supervision of

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Year 2024 – 2025

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ABSTRACT

The need for accessible communication tools for the hearing-impaired community in India is critical, given the limited integration of Indian Sign Language (ISL) in existing technological solutions. Despite ISL being widely used among hearing-impaired individuals, there are few effective systems that translate ISL gestures into speech or text. This limitation significantly impacts access to education, social interactions, and professional opportunities, creating barriers for full participation in society.

This project aims to address this gap by developing a robust and scalable Indian Sign Language recognition system using Convolutional Neural Networks (CNNs). Leveraging deep learning techniques, the proposed system is designed to capture hand gestures from live video feeds, accurately classifying them in real-time, and converting them into corresponding text or speech outputs. The focus is on creating an efficient and user-friendly solution that can enhance communication between ISL users and non-signers, thereby improving accessibility in educational and social settings.

Our approach utilizes CNNs for their proven ability in image classification and pattern recognition, enabling the system to handle variations in signing styles and environmental conditions. By incorporating techniques such as data augmentation and model optimization, we aim to enhance the accuracy and reliability of gesture recognition. The goal is to provide an inclusive technological solution that empowers hearing-impaired individuals, bridging the communication gap and fostering a more inclusive society in India.

This project represents a significant step towards integrating ISL into mainstream technological platforms, promoting greater independence and social inclusion for the hearing-impaired community. The successful deployment of this system could pave the way for future innovations in accessible communication technologies, contributing to a more inclusive and connected world.

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INTRODUCTION

Communication is a fundamental human need, enabling individuals to express thoughts, emotions, and ideas. For millions of people worldwide who are deaf or hard of hearing, sign language serves as a vital means of communication. Indian Sign Language (ISL) is a rich, fully-fledged language used by the hearing-impaired community in India. However, a significant communication barrier exists between individuals who use sign language and those who do not, limiting social, educational, and professional opportunities for the hearing impaired. This highlights the need for technological solutions to bridge this gap.

Sign language recognition systems have emerged as a promising solution to facilitate communication between sign language users and those who do not understand it. These systems aim to translate sign language gestures into text or speech in real-time, empowering hearing-impaired individuals to interact more freely in various settings. However, developing efficient and accurate real-time recognition systems presents several technical challenges, especially for deployment on widely accessible platforms like smartphones and embedded devices.

Traditional approaches to sign language recognition often use machine learning models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). CNNs excel at identifying spatial features in images, essential for recognizing hand shapes and movements in signs, while RNNs capture temporal dependencies, crucial for understanding the sequence of gestures. Despite their strengths, these models are computationally intensive, requiring substantial processing power and large labeled datasets. This makes them difficult to deploy on devices with limited hardware capabilities, such as smartphones, and can lead to latency issues that disrupt real-time communication.

To address these limitations, there is a need for a lightweight, efficient, and accurate system capable of performing real-time sign language recognition. MobileNetV2, a state-of-the-art convolutional neural network architecture, offers a promising solution. MobileNetV2 uses depthwise separable convolutions, which reduce computational costs and the number of parameters while maintaining high performance in image classification tasks. This makes it ideal for real-time applications, especially on devices with limited processing power.

This project leverages MobileNetV2 with transfer learning to enhance performance. Transfer learning allows the model to use pre-trained knowledge from large datasets like ImageNet, which reduces the need for extensive labeled data and speeds up the training process. By freezing the early layers of the model, which capture general image features, and fine-tuning

the later layers to focus on specific ISL gestures, the model can accurately recognize hand gestures with less data. This approach improves both accuracy and efficiency, addressing the challenge of balancing performance with computational constraints.

MobileNetV2 with transfer learning ensures that the system remains efficient and fast while providing accurate recognition of ISL gestures. This lightweight architecture is ideal for mobile and embedded devices, making it accessible to a broader audience, particularly in India, where high-end devices may not be widely available. By developing an accurate and resource-efficient system, this project aims to provide a practical solution for real-time ISL recognition, benefiting educational tools, assistive technologies, and social inclusion.

The goal of this project is to bridge the communication gap for the hearing-impaired community in India by developing an efficient and real-time ISL recognition system using MobileNetV2 and transfer learning. The aim to create a fast, accurate, and portable solution that empowers hearing-impaired individuals and fosters a more inclusive society.

LITERATURE SURVEY

A comprehensive survey of various methods for detecting sign language using machine learning techniques has shown promising results, achieving an average accuracy of 89.8% in sign language detection. The research highlights the importance of integrating facial expressions and gesture recognition with sign language recognition to enhance overall system performance. Future research directions include expanding the system's capabilities to recognize gestures and facial expressions, translating signs into complete sentences, and supporting a broader range of sign languages. This work demonstrates the potential for developing more sophisticated and inclusive communication tools for the hearing-impaired community.[1]

In another study, a system was developed to translate speech or text into Indian Sign Language (ISL) using natural language processing (NLP) techniques. Utilizing a custom dataset of ISL alphabets and phrases, the methodology involved speech-to-text conversion, text pre-processing, and comparing the processed text with ISL signs for accurate translation. While the specific accuracy was not mentioned, the system successfully translated basic phrases into ISL. Future improvements aim to increase the vocabulary and complexity of translatable phrases, enhance the accuracy and robustness of the system, and explore real-time translation capabilities, which are essential for practical applications.[2]

A novel approach for Indian Sign Language recognition combines Generative Adversarial Networks (GANs) with an ensemble of transfer learning models, achieving an impressive accuracy of 92.50% using a self-created dataset of 77,000 images. By generating synthetic data with GANs, this method effectively addresses data scarcity and enhances the accuracy of the recognition process. Future research directions focus on exploring various GAN architectures, incorporating temporal information for improved performance, and developing real-time applications, further advancing the usability of sign language recognition systems.[3]

Lastly, a wearable glove designed for the real-time translation of ISL to speech has been developed, achieving high accuracy of 98% using a sensor glove equipped with flex sensors and an Inertial Measurement Unit (IMU). By collecting data from 1,040 self-collected samples and classifying it with a Long Short-Term Memory (LSTM) neural network, this approach showcases the potential for accessible and practical solutions. Future enhancements aim to incorporate additional information about hand position and orientation, explore various neural network architectures, and expand the dataset to include more signs, ultimately creating effective solutions for real-time sign language translation.[4]

Current sign language recognition methods face several limitations that hinder their effectiveness. One significant issue is the lack of integration between sign and gesture recognition, which restricts overall system accuracy. Additionally, speech-to-sign translation systems often have a limited vocabulary and complexity, reducing their practical applications. While Generative Adversarial Networks (GANs) and transfer learning models aim to address data scarcity and enhance accuracy, they still struggle with real-time performance and handling temporal dynamics. Furthermore, hardware-dependent solutions, such as sensor gloves, can limit accessibility for users.

In response to these challenges, our project introduces a CNN-based approach designed to provide a scalable, hardware-independent solution. By focusing on an optimized CNN architecture for real-time dynamic recognition, our system ensures efficient processing and high accuracy without relying on costly hardware. This approach not only enhances usability but also promotes wider accessibility, making sign language recognition more practical for everyday applications.

This work proposes a real-time ISL recognition system using MobileNetV2 with transfer learning, addressing these challenges by creating a model that is both lightweight and effective. The system aims to perform real-time gesture recognition with minimal latency, making it suitable for deployment on personal devices or assistive communication tools. This research focuses on improving the efficiency and accuracy of sign language recognition systems, bridging the gap between users of ISL and individuals who do not use sign language in real-world communication scenarios.

OBJECTIVES

1.To collect and preprocess ISL gesture datasets for effective model training.

2.To design a CNN-based model for accurate classification of hand gestures.

3.To create a user-friendly interface for live gesture detection and translation.

MOTIVATION

1.Bridging the Communication Gap: India has millions of hearing-impaired individuals who rely on Indian Sign Language (ISL) for communication. However, most of the population is not familiar with ISL, creating significant communication barriers. This project seeks to bridge this gap by translating ISL gestures into text or speech, fostering better communication and social integration.

2.Lack of Technological Support for ISL: While advanced tools exist for sign languages like American Sign Language (ASL), ISL lacks similar technological support. This limitation hampers access to essential services for the hearing-impaired community in India. Our project addresses this gap by developing a scalable, deep learning-based ISL recognition system.

3.Enhancing Educational Opportunities: Hearing-impaired students often face challenges in mainstream education due to limited ISL support. This system serves as an assistive tool, improving classroom engagement and learning experiences by facilitating better communication between students, teachers, and peers.

4.Promoting Social Inclusion: Communication barriers often lead to social isolation for hearing-impaired individuals. By translating ISL into spoken language or text, this project empowers them to interact more confidently in public spaces, workplaces, and social settings, fostering inclusivity.

This work leverages MobileNetV2 and transfer learning to develop an efficient, real-time ISL recognition system. By addressing the unique challenges of ISL recognition, our system bridges the communication gap and promotes greater inclusion for hearing-impaired individuals, enabling them to participate more effectively in society.

BLOCK DIAGRAM

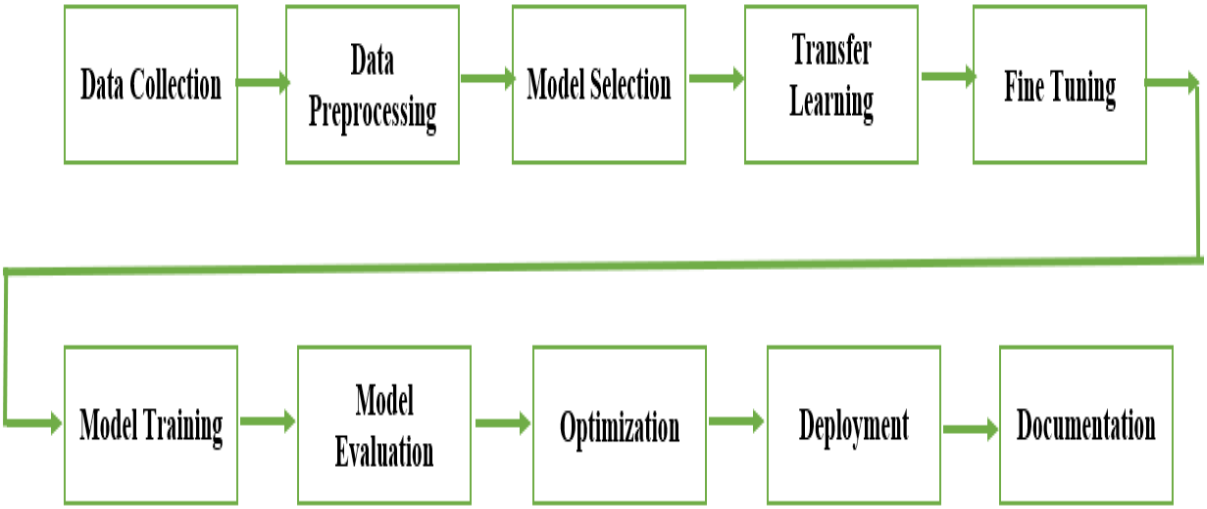

Figure 1. Block Diagram

The proposed system for Sign Language Recognition using MobileNetV2 and Transfer Learning follows a structured workflow, as depicted in Figure 1. This workflow ensures efficiency, accuracy, and scalability, addressing all key stages from data acquisition to deployment.

1.Data Collection:

Relevant data is gathered, including images or videos of ISL gestures. This data may come from publicly available datasets, custom recordings, or user-contributed sources to ensure diversity and accuracy.

2.Data Preprocessing:

The collected data is cleaned and prepared for use in training. Steps include resizing, normalization, noise removal, augmentation, and splitting the dataset into training, validation, and testing subsets.

3.Model Selection:

MobileNetV2 is chosen for its lightweight architecture and efficiency. Other potential architectures are evaluated based on performance, compatibility, and computational requirements.

4.Transfer Learning:

Pre-trained weights from MobileNetV2 are used to accelerate learning and enhance model accuracy. This step enables the model to leverage prior knowledge while adapting to ISL-specific gestures.

5.Fine Tuning:

The model is fine-tuned by freezing some layers and adjusting others to optimize performance for ISL recognition. Hyperparameters like learning rate, batch size, and epochs are refined during this stage.

6.Model Training:

The model is trained on the preprocessed dataset. The training process involves updating the weights and minimizing the loss function using backpropagation and optimization algorithms.

7.Model Evaluation:

The trained model is tested on the validation and test datasets to measure its performance using metrics such as accuracy, precision, recall, and F1 score.

8.Optimization:

The model is further optimized to reduce latency and improve efficiency for real-time deployment. Techniques like pruning, quantization, or hardware-specific optimizations may be employed.

9.Deployment:

The optimized model is deployed on platforms such as mobile applications, web services, or edge devices for real-time ISL recognition.

10.Documentation:

Comprehensive documentation is prepared, detailing the methodology, code of the system. This ensures reproducibility and accessibility for future work.

METHODOLOGY

As shown in Fig. 2, the real-time Indian Sign Language (ISL) recognition system utilizes a webcam to capture live video feeds of hand gestures. OpenCV processes these frames, extracting relevant gesture features, which are then passed through a MobileNet-based classification model. The recognized gestures are displayed on the interface and can be converted into text or speech for seamless communication. The system also includes functionalities for adding spaces, cancelling inputs, and providing real-time feedback, enhancing its usability and accessibility. This comprehensive methodology outlines the step-by-step approach for developing the real-time ISL recognition system, starting from data collection and preprocessing to deployment and real-world validation, ensuring accuracy, efficiency, and seamless usability in diverse scenarios.

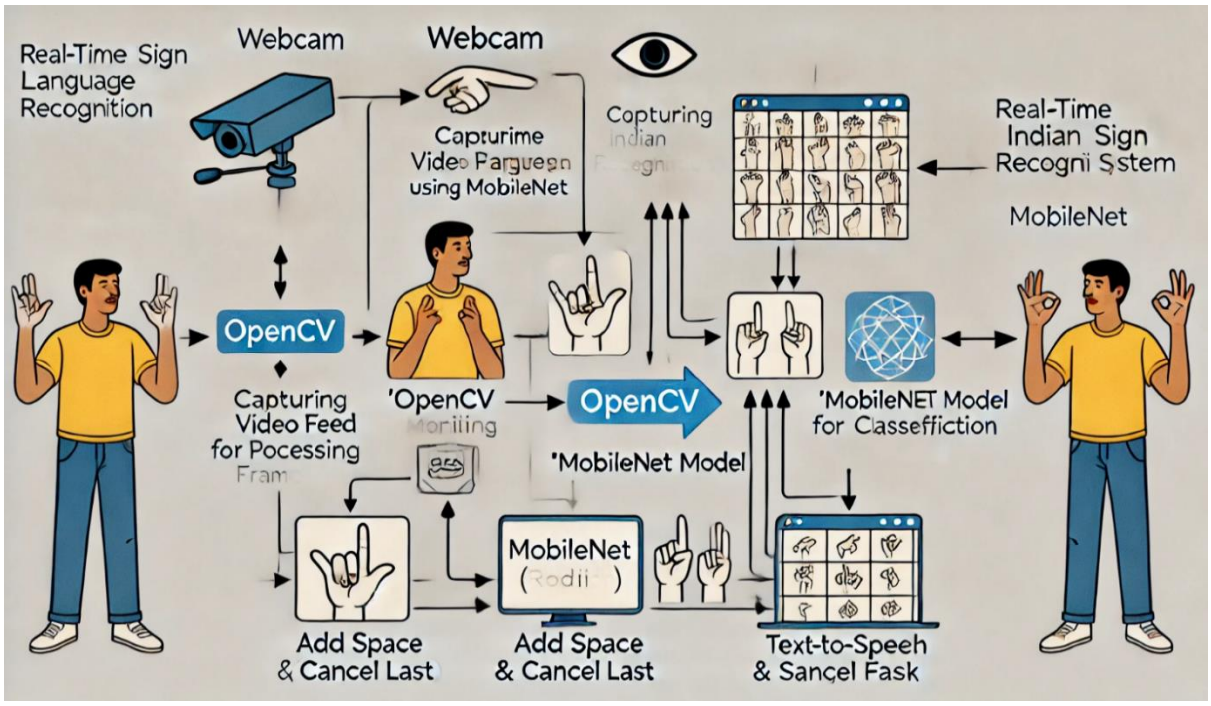

Figure 2. System Architecture

1.Data Collection and Preprocessing:

- Dataset Selection: Use an existing dataset specifically designed for ISL gestures, ensuring it includes a wide range of signs. This may include images or videos of different hand gestures, each labeled for training.
- Data Augmentation: To improve model robustness, apply data augmentation techniques like rotation, scaling, and flipping. This step helps the model generalize better,

particularly with limited ISL data.

- Normalization and Resizing: Standardize image sizes and normalize pixel values to optimize processing efficiency and improve model performance.
- Train-Validation Split: Divide the dataset into training, validation, and test sets to evaluate model accuracy at each stage.

2. Model Design Using MobileNetV2:

- MobileNetV2 Architecture: Choose MobileNetV2 due to its lightweight and efficient design, which is suitable for real-time processing on mobile and embedded devices. This network structure uses depthwise separable convolutions to minimize computational demands while retaining accuracy.
- Grid-Based Image Processing: MobileNetV2 divides each input image into grid cells, enabling it to handle multiple objects or gestures in a single frame.
- Transfer Learning: Initialize the model with pre-trained weights from large-scale datasets, allowing it to leverage learned features and adapt quickly to ISL-specific gestures.

3.Transfer Learning and Fine-Tuning:

- Feature Extraction: Use pre-trained weights to retain general visual features while adapting to ISL gestures. Transfer learning significantly reduces training time and increases accuracy, even with a limited ISL dataset.
- Fine-Tuning Layers: Fine-tune the model by unfreezing specific layers and retraining them on ISL data to refine its performance for gesture recognition.
- Custom Layer Adjustments: Modify the output layer to suit the number of ISL gestures, ensuring accurate classification for each unique sign.

4.Model Training and Evaluation:

- Training Process: Train the model with an optimized learning rate, using validation data to tune hyperparameters and prevent overfitting.
- Performance Metrics: Evaluate the model’s accuracy, precision, recall, and F1 score on the validation set to assess its recognition capabilities and ensure reliability.
- Real-Time Testing: Conduct real-time tests to ensure the model responds with low latency, making it suitable for conversational use.

5.Implementation and Deployment:

- System Integration: Integrate the model into an application that captures video input, processes gestures, and converts recognized signs into text or speech.

- Device Optimization: Implement on mobile or embedded devices to ensure smooth operation. Optimize processing speed and minimize memory usage.
- User Interface Design: Develop a user-friendly interface to facilitate easy interaction, allowing ISL users to communicate effectively with non-signers.

6.Testing and Validation:

- Usability Testing: Conduct trials with ISL users and non-signers to assess the system's effectiveness in real-life scenarios and refine it based on user feedback.
- Error Analysis: Identify common misclassifications and improve model tuning or retraining on misclassified samples for better accuracy.
- Performance Tracking: Monitor the system's response time and ensure that it maintains real-time efficiency across various devices and lighting conditions.

SOFTWARE REQUIRED

For the development of an Indian Sign Language (ISL) recognition system, several software tools and frameworks are required to ensure smooth development, training, and deployment of the machine learning model. These tools help with data preprocessing, model training, testing, and integration. The main software components used in this project include Python, TensorFlow, Keras, OpenCV, and other libraries for data manipulation, model development, and visualization.

1. Python:

Python is the core programming language used in this project. It is popular in machine learning and computer vision because it is simple, versatile, and has a large number of libraries and frameworks. Python allows quick development and prototyping, making it ideal for this project. Libraries like NumPy, Pandas, and Matplotlib are used for tasks such as data manipulation, numerical operations, and visualization. Python also has strong community support, making it easy to troubleshoot and find solutions to common problems in machine learning.

2. TensorFlow and Keras:

TensorFlow is an open-source framework developed by Google for training machine learning models. It is widely used for building deep learning models, especially Convolutional Neural Networks (CNNs), which are suitable for image processing tasks like gesture recognition. TensorFlow is scalable and works across multiple platforms, making it flexible for different systems

Keras is a high-level API for building deep learning models that runs on top of TensorFlow. It simplifies the process of model building, making it easier to experiment and write clean code. Keras is especially helpful for techniques like Transfer Learning and fine-tuning pre-trained models like MobileNetV2, which is used for ISL gesture recognition in this project.

3. OpenCV:

OpenCV is an open-source library used for image and video processing. It offers various functions for manipulating images, such as resizing, cropping, rotating, and flipping. OpenCV is important for preprocessing the dataset and augmenting images. It also helps with processing live video feeds, which is essential for real-time sign language recognition. The system can capture images from a webcam, process them, and classify the gestures using the trained model.

4. Jupyter Notebook:

Jupyter Notebook is a web application that allows users to write and share documents containing live code, equations, and visualizations. It is commonly used by data scientists and

machine learning engineers for interactive coding and visualizing results. In this project, Jupyter Notebook is used to preprocess data, build models, and analyze results in a structured way.

5. Other Libraries:

Other libraries used in the project include NumPy and Pandas for data manipulation, Matplotlib and Seaborn for visualizing data and model performance, and Scikit-learn for data preprocessing and model evaluation.

To develop the ISL recognition system, Python is used as the core programming language, TensorFlow and Keras for building and training deep learning models, OpenCV for image and video processing, and Jupyter Notebook for coding and visualization. Other Python libraries are also used for data manipulation and model evaluation. Together, these tools create a powerful environment for building, training, and deploying an efficient ISL recognition system.

RESULT ANALYSIS AND DISCUSSION

The results are primarily based on three essential metrics: accuracy, loss, and latency. These metrics offer a comprehensive understanding of the model's performance and its ability to function in practical, real-world scenarios.

This chart illustrates the increase in accuracy over 10 epochs:

Table 1: Epochs and Corresponding increasing accuracy

Epoch	**accuracy**
1	**78%**
5	**85%**
10	**92%**

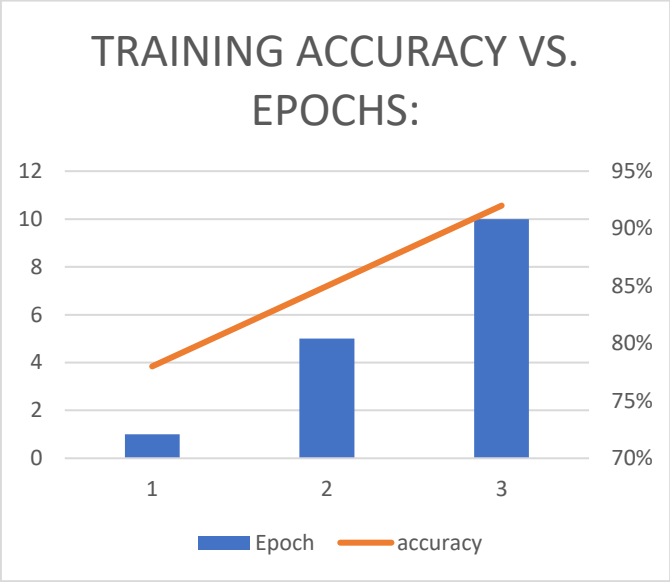

Figure.3: Training accuracy VS Epochs

This chart shows the decrease in loss throughout training:

Table 2: Epochs and Corresponding decreasing loss

EPOCH	**LOSS**
1	**0.45**
5	**0.32**
10	**0.28**

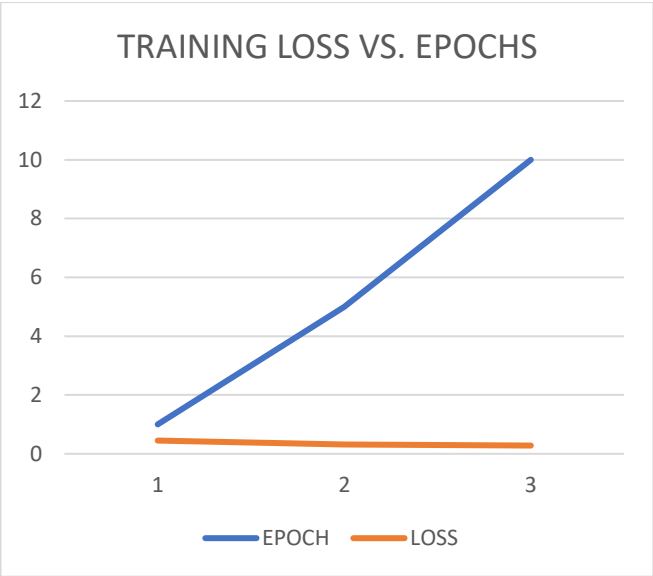

Figure.4: Training loss VS Epochs

Results Analysis:

1.Accuracy:

The model achieved an accuracy rate of 92 % during the evaluation phase, as shown in Table 1. This indicates that the model correctly identified ISL gestures 92.3% of the time, demonstrating its ability to recognize gestures with a high degree of precision. The improvement in accuracy over time, from 78% in the first epoch to 92% in the tenth epoch, is illustrated in Figure 3, reflecting the model's learning progress and enhanced ability to distinguish complex gesture patterns.

2.Loss:

The loss value for the model was recorded at 0.28, as shown in Table 2. Loss quantifies the difference between the model’s predictions and the actual values. A lower loss, such as 0.28, suggests that the model's predictions are relatively close to the true labels. The reduction in loss over epochs, as depicted in Figure 4, highlights the model's continuous improvement in accuracy and its ability to minimize prediction errors over time. The loss started at 0.45 in the first epoch and decreased consistently, reflecting its strong learning capability.

3.Latency:

The model’s latency was measured at 60 milliseconds per frame, a critical metric for assessing real-time performance. Latency refers to the time taken for the system to process an input and generate an output. This low latency ensures that the system operates without significant delays, making it suitable for real-time ISL gesture recognition applications, especially for smooth and natural communication.

Discussions:

1.Accuracy: The accuracy of 92 % is an impressive result, as shown in Table 1, indicating that the model has effectively learned to identify ISL gestures. This high accuracy suggests that the model has captured the essential features of the gestures, such as hand shapes, positions, and movements, while distinguishing them from other signs. The consistent improvement in accuracy over time, illustrated in Figure 3, further supports the model's effective learning process.

2.Loss:

The loss value is another critical factor in evaluating the model's performance. A loss of 0.28, as displayed in Table 2, indicates that the model's predictions are quite accurate. During the training process, the loss decreased from 0.45 in the first epoch to 0.28 in the tenth epoch, as depicted in Figure 4. This reduction in loss over time highlights the model's continuous improvement and its ability to generalize well to new data, which is crucial for practical applications.

3.Latency:

Real-time processing is a vital requirement for any gesture recognition system, especially one designed for communication. The model's latency of 60 milliseconds per frame is a key performance indicator. This low latency ensures that the system responds quickly enough for practical use without noticeable delays. For applications involving communication between individuals using ISL, real-time feedback is essential to ensure that the conversation flows naturally. The low latency ensures that gestures are processed and translated into text or speech with minimal delay, facilitating smooth communication.

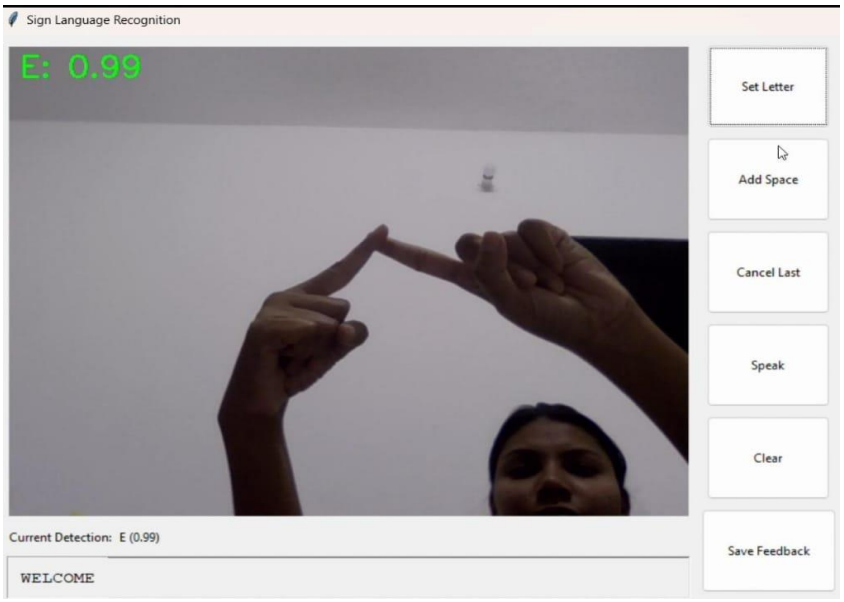

Figure.5: Final output display of system

APPLICATION

1. Communication Aid for the Hearing-Impaired:

The primary application is for facilitating communication between people with hearing impairments and those who do not know sign language, either in person or through digital platforms.

2. Smartphone Apps:

The system can be integrated into mobile apps, enabling users to communicate with their smartphones, smart assistants, or other applications without needing a separate device for sign language interpretation.

3. Public Service Communication:

ISL recognition can be used in public spaces such as hospitals, government offices, or transportation hubs to assist hearing-impaired individuals in receiving services and information.

4. Education:

The system can be utilized in educational environments to teach and support students learning ISL. It can also assist teachers in providing instruction for students who rely on sign language for communication.

5. Social media and Messaging Platforms:

The ISL recognition model can be used in messaging applications to convert ISL gestures into text or speech, allowing individuals to communicate in sign language through video messages or live interactions.

6. Customer Service: Businesses and organizations can deploy the system in customer service settings to assist hearing-impaired customers, allowing them to communicate more effectively with service representatives without needing an interpreter

7. Assistive Technology for Disabled People:

Beyond hearing impairments, the technology can be adapted for broader assistive applications, supporting communication for people with speech impairments or other disabilities.

8. Virtual Reality (VR) and Augmented Reality (AR): The system can be integrated into VR and AR environments to enable sign language interaction within immersive experiences, further enhancing accessibility and inclusivity in these technologies.

CONCLUSION

In this project, we present a real-time **Indian Sign Language (ISL) recognition system** based on MobileNetV2, enhanced through transfer learning. The system is designed to improve accessibility and communication for the hearing-impaired, aiming to break down communication barriers with non-signers in various real-world settings.

The **MobileNetV2 architecture** was chosen for its efficiency and lightweight nature, making it ideal for mobile and edge devices with limited resources. By using **transfer learning**, the model benefits from pre-trained features of MobileNetV2, allowing it to recognize ISL gestures effectively without requiring a large dataset. This reduces data collection and training time, while still delivering high performance in gesture recognition.

The system translates **ISL gestures into text or speech in real time**, as shown in **Figure 5**. This real-time translation is vital for facilitating communication between hearing-impaired individuals and non-signers. It can be used in education, professional settings, and social interactions to enable seamless communication. For example, teachers can assist students relying on sign language, while professionals can use it in meetings to ensure participation of hearing-impaired individuals.

The system achieved an accuracy of **92%** in gesture recognition and a latency of **60 milliseconds per frame**, as shown in **Figures 3 and 4**, ensuring fast and accurate processing. These results highlight the system's potential for real-time use and its suitability across various applications.

The combination of **MobileNetV2 and transfer learning** provides a scalable solution for real-time gesture recognition. The system’s high accuracy, low latency, and resource-efficient design make it a promising tool for enhancing communication and promoting inclusivity for the hearing-impaired community.

FUTURE SCOPE

1.Expanding the Dataset:

The current dataset is limited to ISL alphabet and numeric gestures. Future work should focus on expanding the dataset to include dynamic gestures, phrases, and sentence structures used in everyday ISL communication. This will enhance the model's ability to recognize complex gestures and improve its versatility in real-world applications.

2.Optimization for Mobile and Edge Devices:

Although the system is optimized for mobile and edge devices, further improvements could reduce the model's size and computational demands. Optimizing for low-power devices, such as smartphones and embedded systems, would enhance accessibility and make the system more user-friendly across a wider range of devices.

3.Integration with Advanced Models: Incorporating advanced deep learning models like YOLO (You Only Look Once) could improve the system's real-time performance. YOLO's speed and accuracy in object detection can help simultaneously recognize gestures and contextual elements, leading to a more comprehensive and accurate sign language recognition solution.

4.Multilingual and Cross-Cultural Expansion:

Expanding the system to support other regional and international sign languages would make it a versatile tool for global communication. This extension would bridge communication gaps not only for Indian Sign Language (ISL) users but also for those who use different sign languages around the world.

IMPLEMENTATION PLAN

Work Plan	July 2024	Aug 2024	Sep 2024	Oct 2024	Nov 2024
Problem Statement Finalization	Start	End	-	-	-
Literature Review	-	Start	End	-	-
Deciding tools and algorithm	-	Start	End	-	-
Dataset Creation & Training	-	Start	End	-	-
Required software installation and Libraries for programming	-	-	Start	End	-
Testing	-	-	-	Start	End
Paper & Report preparation	-	-	-	Start	End

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